

Lab Sheet # L2

Instructions:

- Submit *.sql file, named as Lab02_6XXXXXX.sql to LAs or instructors prior to My Courses.
- If students cannot finish within the class time, students can submit until the end of the next class.

Self-study:

1. SELECT

<https://www.mysqltutorial.org/mysql-basics/mysql-select-from/>

<https://www.mysqltutorial.org/mysql-aggregate-functions/>

2. JOIN

<https://www.mysqltutorial.org/mysql-basics/mysql-join/>

3. ROUND

<https://www.mysqltutorial.org/mysql-math-functions/mysql-round-function/>

Today's Task

1. Database Setup:

Create a database named **L02_classicmodels_db** by executing the sql-file.

2. How many customers are located in NYC?

Expected 1 row, 1 column.

NYC Customers
5

3. What is the average amount of the payment and the standard deviation of the payments occurred in the third quarter of the year?

Note: Third quarter of every year is from 1 July – 30 September

Expected 1 row, 2 columns.

average_payment	payment_std_dev
30988.898182	21070.206013162853

4. Show the number of offices located in each country and state respectively, ordered by country names.

Expected 7 rows, 3 columns.

country	state	numOffices
Australia	HULL	1
France	HULL	1
Japan	Chiyoda-Ku	1
UK	HULL	1
USA	CA	1
USA	MA	1
USA	NY	1

5. Show the list of the product lines that have the number of the products at least 10 but not more than 20 products. The result must include the name of the product line, the total numbers of the products and the range of MSRP per product line, sorted by the name of product line.

Note: The range of MSRP is calculated from the difference of the maximum and the minimum MSRP value

Expected 2 rows, 3 columns.

productLine	numProd	rangeMSRP
Motorcycles	13	153.43
Planes	12	108.03
Trucks and Buses	11	82.56

6. Calculate the average of the price difference between 'MSRP' and 'buyPrice' of each product line. The result must include the name of the product line and its average difference price, sorted from the highest to lowest difference value.

Expected 7 rows, 2 columns.

productLine	avgDiff
Classic Cars	53.57
Trucks and Buses	46.85
Motorcycles	46.49
Vintage Cars	41.03
Planes	39.89
Ships	39.56
Trains	29.93

7. How many “limited” company customers that are under the responsibility of the sale representative named, “Leslie Jennings”

Note: 'Ltd.' used in the name of a company indicates a limited company

Expected 1 row, 1 column.

numLtdComp
2

8. List all sale managers (employee number, full name, job title) and their subordinate sale representatives under their supervision.

Note: The attribute 'reportsTo' shows the recursive relationship. The value indicates the employee number of their manager that he/she reports to.

Expected 3 rows, 4 columns.

employeeNumber	fullname	jobtitle	numSubEmp
1088	William Patterson	Sales Manager (APAC)	3
1102	Gerard Bondur	Sale Manager (EMEA)	6
1143	Anthony Bow	Sales Manager (NA)	6

9. List the products (code and name) that have never been sold.

Expected 1 row, 2 columns.

productCode	productName
S18_3233	1985 Toyota Supra

10. List all combinations of the “sales representative” employees who work in USA office and all US customers. Shows Sales Representatives, their locations, Customer names, and Customer locations.

Hint: There are 36 US customers and 6 sales representatives located in USA

Expected 216 rows, 4 columns.

SalesRep	SalesLocation	customerName	CustomerLocation
George Vanauf	NYC, NY	Diecast Classics Inc.	Allentown, PA
Foon Yue Tseng	NYC, NY	Diecast Classics Inc.	Allentown, PA
Steve Patterson	Boston, MA	Diecast Classics Inc.	Allentown, PA
Julie Firrelli	Boston, MA	Diecast Classics Inc.	Allentown, PA
Leslie Thompson	San Francisco, CA	Diecast Classics Inc.	Allentown, PA
Leslie Jennings	San Francisco, CA	Diecast Classics Inc.	Allentown, PA
George Vanauf	NYC, NY	Gifts4AllAges.com	Boston, MA
Foon Yue Tseng	NYC, NY	Gifts4AllAges.com	Boston, MA

11. List all contact names and phone numbers of both customers and the employees in one result.

- Contact name of customers is derived from customers' contact firstname and lastname
- Phone number of employees is combined from the office phone number and extension with dash, e.g., [phone number]-[extension]

Expected 145 rows, 3 columns.

contact_name	phone_number	contact_type
Adrian Huxley	+61 2 9495 8555	Customer
Akiko Shimamura	+81 3 3584 0555	Customer
Alejandra Camino	(91) 745 6555	Customer
Alexander Feuer	0342-555176	Customer
Alexander Semenov	+7 812 293 0521	Customer
Allen Nelson	6175558555	Customer
Andy Fixter	+61 2 9264 2451-x101	Employee
Ann Brown	(171) 555-0297	Customer
Anna O'Hara	02 9936 8555	Customer

12. Select top 3 of the most sold product (judged by the total numbers of quantity ordered) in the following product lines: Motorcycles, Classic Cars, and Vintage Cars respectively.

Expected 9 rows, 4 columns rows.

productLine	productCode	productName	total_quantity
Classic Cars	S18_3232	1992 Ferrari 360 Spider red	1808
Classic Cars	S24_3856	1956 Porsche 356A Coupe	1052
Classic Cars	S10_4757	1972 Alfa Romeo GTA	1030
Motorcycles	S10_1678	1969 Harley Davidson Ulti...	1057
Motorcycles	S24_1578	1997 BMW R 1100 S	1033
Motorcycles	S12_2823	2002 Suzuki XREO	1028
Vintage Cars	S18_1342	1937 Lincoln Berline	1111
Vintage Cars	S18_3856	1941 Chevrolet Special De...	1076
Vintage Cars	S50_1341	1930 Buick Marquette Pha...	1074